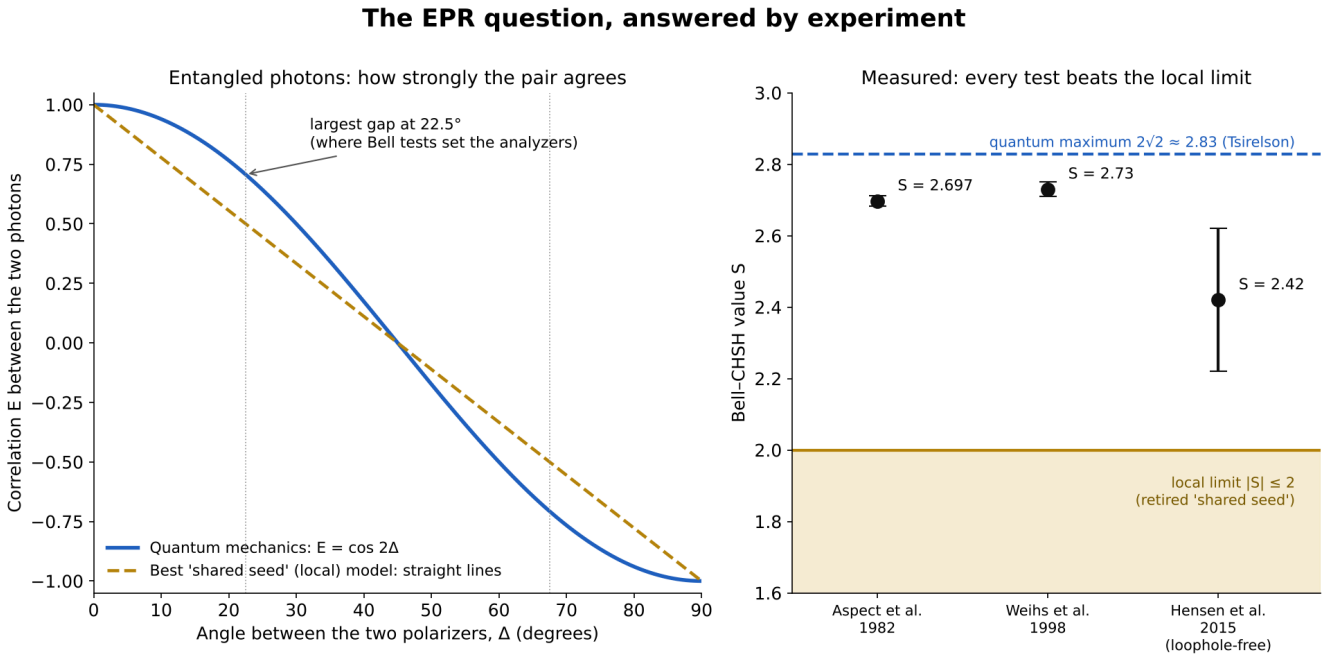
The EPR Paradox: the verdict

Shelf: Established. Uniquea archive · github.com/phantomovanish/Uniquea · You are welcome to print this page, or upload it to any AI assistant and ask your own questions.



EPR (1935) asked whether particles carry hidden instructions. Bell (1964) turned that into a number S: hidden instructions can never exceed 2. Experiments find S above 2, as quantum mechanics predicts. Shelf: Established. Archive: the shared-seed idea is retired (Master §2.4, §3.11).

The question (1935)

Einstein, Podolsky and Rosen asked whether two particles from one source carry hidden instructions, a "shared seed," that fix every result in advance. If so, quantum mechanics would be incomplete.

How it became testable (1964)

John Bell turned the question into a number, S. Any model with hidden instructions and no faster-than-light influence (a *local hidden-variable* model) must give **|S| ≤ 2**. Quantum mechanics predicts up to **2√2 ≈ 2.83** (Tsirelson's bound).

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|ψ⁻⟩ = ( |↑↓⟩ − |↓↑⟩ ) / √2
E(a, b) = −cos θ              (spin-½)        E(α, β) = −cos 2(α − β)   (photons)
|S| ≤ 2  under Bell's assumptions             |S| ≤ 2√2  in quantum mechanics
P(A = ±1 | a, b) = ½   for every b            (no signalling)
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The verdict (experiments)

Experiment	Measured S	Note
Aspect, Grangier and Roger, 1982	2.697 ± 0.015	Photons, beyond the local limit
Weihs et al., 1998	2.73 ± 0.02	Settings chosen fast and far apart
Hensen et al., 2015 (Delft)	2.42 ± 0.20	First loophole-free test

Several loophole-free tests followed in 2015. The Nobel Prize in Physics 2022 went to John Clauser, Alain Aspect and Anton Zeilinger for this work.

Verdict: local hidden-variable models, the "shared seed," are ruled out. No signal travels between the particles: each side alone sees 50/50 results, whatever the other side does. How to *interpret* entanglement remains open.

What Uniquea retired, and what it keeps open

Claim in the archive	Label
Entanglement is a shared seed or "DNA script" from the common source	**Contradicted.** A local hidden-variable model; simulated, |S| = 2.000 against the quantum 2.828
360/127 ≈ 2√2	Coincidence (0.22% apart). The Tsirelson bound is exact
EPR Double-Fulcrum: one bilateral state, two measurements	Candidate interpretation; standard quantum mechanics relabelled, no new prediction
A field wake carries the entanglement from A to B	**Withdrawn** (it would allow signalling)

Open target: derive the 2√2 cap from the structure of the plenum.

The retirement is a design requirement for Uniquea: its quantum layer cannot be local hidden variables.

Two notes for reusing the figure

- With S = E(a,b) + E(a,b′) + E(a′,b) − E(a′,b′), photon settings a = 45°, a′ = 0°, b = 22.5°, b′ = 67.5° give −2√2. With the minus sign on E(a,b′), use a = 0°, a′ = 45°.
- For spin-½ the Bloch-sphere angle equals the lab angle; for photons the Poincaré-sphere angle is twice the lab angle. That is the whole factor of 2.

Source in the archive: Master §2.4 and §3.11 (master/Uniquea-Master-Public.md).